Technical Report: iOS Application "Gnome"

1. Project Structure

The project is a native iOS application developed using the **Swift** language and the **SwiftUI** declarative framework. The architecture follows the modern *SwiftUI App* lifecycle, integrating components from *UIKit* and *MapKit* where necessary.

The file organization and their responsibilities are as follows:

A. Entry Point and Configuration

- **gnomeApp.swift**: This is the root file of the project. It contains the main structure marked with @main, which defines the entry point of the application. Its function is to instantiate the main window (WindowGroup) and launch the initial view, ContentView.

B. Core Logic and Interface

- **ContentView.swift**: This is the most critical and extensive file in the project. It acts as the main controller and container for the user interface. Its responsibilities include:
  - **State Management (@State):** It manages global session variables, such as the list of completed gnomes (completedGnomes), captured images (gnomeImages), and favorites (favorites).
  - **Navigation:** It implements a TabView that divides the app into three main sections: Map, Gallery, and Favorites.
  - **Data Model:** It defines the struct Gnome, which contains the ID, geographic coordinates, and description for each gnome. It includes a static array with data for approximately 60 real gnomes located in Wrocław.
  - **Map Integration:** It configures the MapKit view, managing the geographic region and annotations (markers) on the map.
  - **Playback Logic:** It manages the LoopingPlayerView to display the introductory video.

C. Reusable View Components

- **GalleryView.swift**: This is a modular component designed to display the gnome collection in a grid format. It receives data and images from the main view and decides whether to display the captured photo or a gray placeholder if the gnome has not yet been found.
- **ImagePicker.swift**: This file acts as a bridge between *SwiftUI* and *UIKit*. It uses the UIViewControllerRepresentableprotocol to wrap the native iOS

UIImagePickerController, allowing the application to access the device's camera or photo library to capture the gnomes.

D. Multimedia Resources (Assets)

- **video-begin.mov**: A video file showing a 3D animation of a gnome winking. It is used for the immersive introduction upon opening the app.
- **IMG_3032-2.png**: A static image of a gnome, likely used as a visual reference or icon within the interface.

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2. Project Description and Functionality

General Concept

The application is a gamified tourism tool focused on the city of Wrocław (Poland). The objective is to guide the user to find the famous gnome statues scattered throughout the city, allowing them to keep a visual record of their findings through photographs.

Workflow

1. Introduction (Onboarding): Upon starting the application, the user is greeted by an introductory screen (showIntro) that plays a looping video (video-begin.mov) of an animated gnome. Once the video ends or is dismissed, the application automatically transitions to the main interface.

2. Exploration (Map Tab): This is the primary view. The user interacts with a map of Wrocław, initially centered on coordinates 51.1079, 17.0385.

- **Markers:** The map displays pins for each gnome. These are color-coded: **Red** indicates the gnome has not been found, and **Green** indicates it has been captured/completed.
- **Capture:** Tapping a marker opens a detail sheet. If the gnome has no photo, it displays its description and an "Add Photo" button. This launches the camera or gallery (via ImagePicker) to save the finding.
- **Gnome of the Day:** There is a floating button that suggests a random "Gnome of the Day" to incentivize exploration.

3. Collection (Gallery Tab): Here, the user can visualize their global progress. A grid with numbered slots from 1 to 64 is presented.

- Slots display the photo taken by the user if the gnome has been found.
- If not found, it shows a gray silhouette.
- From this view, the user can tap a photo to view it in full size, replace it, or delete it.

4. Favorites (Favorites Tab): Allows the user quick access to gnomes they have marked with a "heart." Only gnomes that already have an associated photo and have been explicitly marked as favorites appear in this list.

Notable Technical Features

- **Search:** The application includes a search bar allowing filtering of gnomes by their ID or description text, both on the map and in the gallery.
- **Persistence (Session State):** Currently, progress (photos and completion status) is maintained in memory during the active session via state variables.
- **Touch Interaction:** The map supports gestures, such as double-tapping to zoom into a specific region and saving navigation history to allow backtracking.